

US Healthcare Patient Readmission Analysis

An End-to-End Data Analytics Project

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1. Introduction

- Hospital readmission is one of the most critical and costly challenges in modern healthcare
- In the US, readmissions cost the healthcare system billions of dollars annually
- Diabetic patients are among the highest risk groups for hospital readmission
- This project analyzes real patient data from 130 US hospitals collected between 1999 and 2008
- The goal is to identify which patient groups, clinical factors, and treatment patterns are most associated with readmission
- Insights from this analysis can help hospitals allocate resources better and intervene early for high risk patients

2. Dataset Overview

- Source: UCI Machine Learning Repository / Kaggle
- Dataset name: Diabetes 130-US Hospitals for Years 1999-2008
- Original size: 101,766 rows and 50 columns
- Contains patient demographics, admission details, diagnoses, medications, and readmission status
- Key columns: age, race, gender, time_in_hospital, num_medications, diag_1, diag_2, diag_3, readmitted
- Missing values were encoded as ? instead of standard NaN

encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	...	citaglipton	insulin	glyburide- metformin	glipizide- metformin	glimepiride- pioglitazone	metformin- rosiglitazone	metformin- pioglitazone	change	diabetesMed	readmitted
0	2278392	8222457	Caucasian	Female	[0-10)	?	6	25	1	X	1	...	No	No	No	No	No	No	No	NO
1	149190	55629189	Caucasian	Female	[10-20)	?	1	1	7	3	...	No	Up	No	No	No	No	Ch	Yes	>30
2	64410	86047875	AfricanAmerican	Female	[20-30)	?	1	1	7	2	...	No	No	No	No	No	No	No	Yes	NO
3	500364	82442376	Caucasian	Male	[30-40)	?	1	1	7	2	...	No	Up	No	No	No	No	Ch	Yes	NO
4	16680	42519267	Caucasian	Male	[40-50)	?	1	1	7	1	...	No	Steady	No	No	No	No	Ch	Yes	NO

5 rows x 50 columns

... Shape: (101766, 50)

3. Data Cleaning (Phase 1)

- Loaded dataset using Pandas and performed initial inspection
- Replaced all ? placeholder values with NaN for proper missing value handling
- Identified 9 columns with missing values
- Dropped 5 columns with more than 39% missing data: weight (96.8%), max_glu_serum (94.7%), A1Cresult (83.3%), medical_specialty (49%), payer_code (39.6%)

```
Missing Count  Missing %
weight          98569    96.858479
max_glu_serum   96420    94.746772
A1Cresult       84748    83.277322
medical_specialty 49949    49.082208
payer_code       40256    39.557416
race            2273     2.233555
diag_3           1423     1.398306
diag_2           358     0.351787
diag_1           21      0.020636
```

```
*** Columns after drop: (101766, 45)
```

- Filled race column nulls with 'Unknown' as only 2.2% were missing

```
*** Final shape: (100244, 45)
Overall null sum : 0
```

- Dropped rows with missing diagnosis values (diag_1, diag_2, diag_3) as less than 1.5% were affected
- Removed 3 rows with invalid gender entries
- Dropped irrelevant ID columns: encounter_id and patient_nbr

```
*** Shape after dropping IDs: (100244, 43)
```

- Created two new binary columns: readmitted_binary (1 if readmitted within 30 days) and readmitted_any (1 if readmitted at all)

```
readmitted  readmitted_binary  readmitted_any
1          >30                  0                  1
2          NO                   0                  0
3          NO                   0                  0
4          NO                   0                  0
5          >30                  0                  1
6          NO                   0                  0
7          >30                  0                  1
8          NO                   0                  0
9          NO                   0                  0
10         >30                  0                  1
```

- Final clean dataset: 100,241 rows and 45 columns.

```
Clean file saved successfully!  
Final shape: (100241, 45)
```

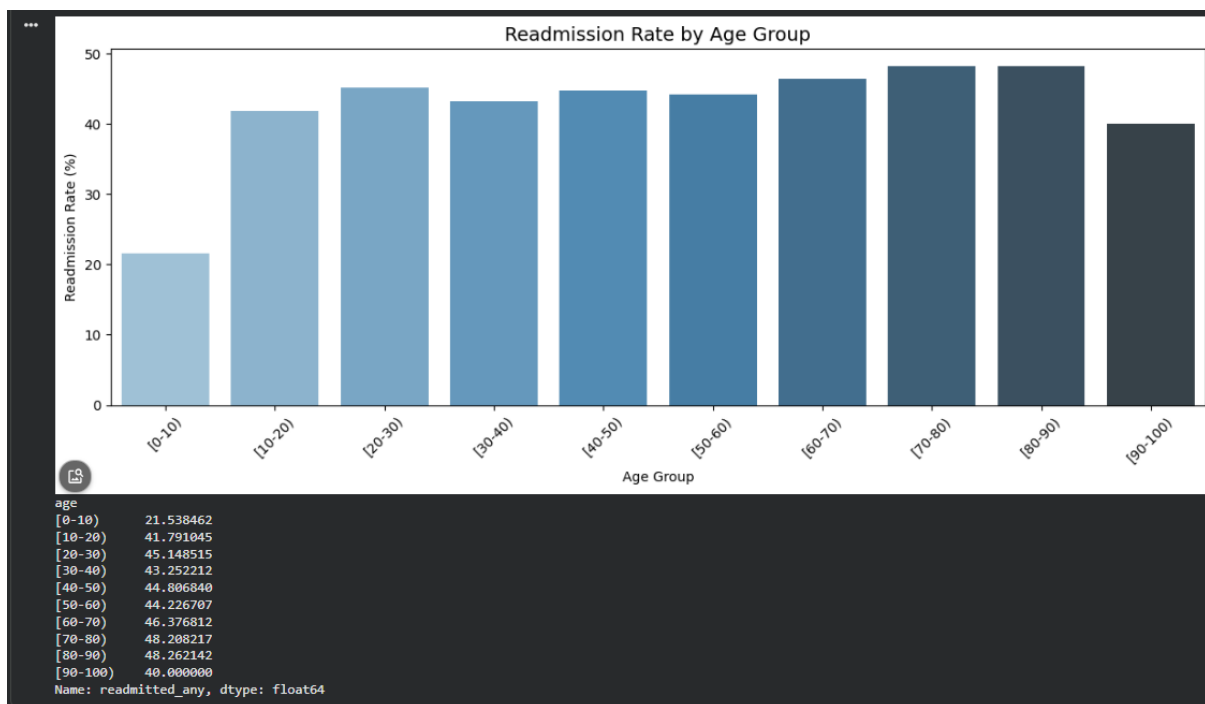
4. Exploratory Data Analysis (Phase 2)

4.1 Readmission Overview

- Overall readmission rate across all patients is 46.31%
- 53,818 patients were not readmitted (53.7%)
- 35,173 were readmitted after 30 days (35.1%)
- 11,250 were readmitted within 30 days — classified as high risk (11.2%)

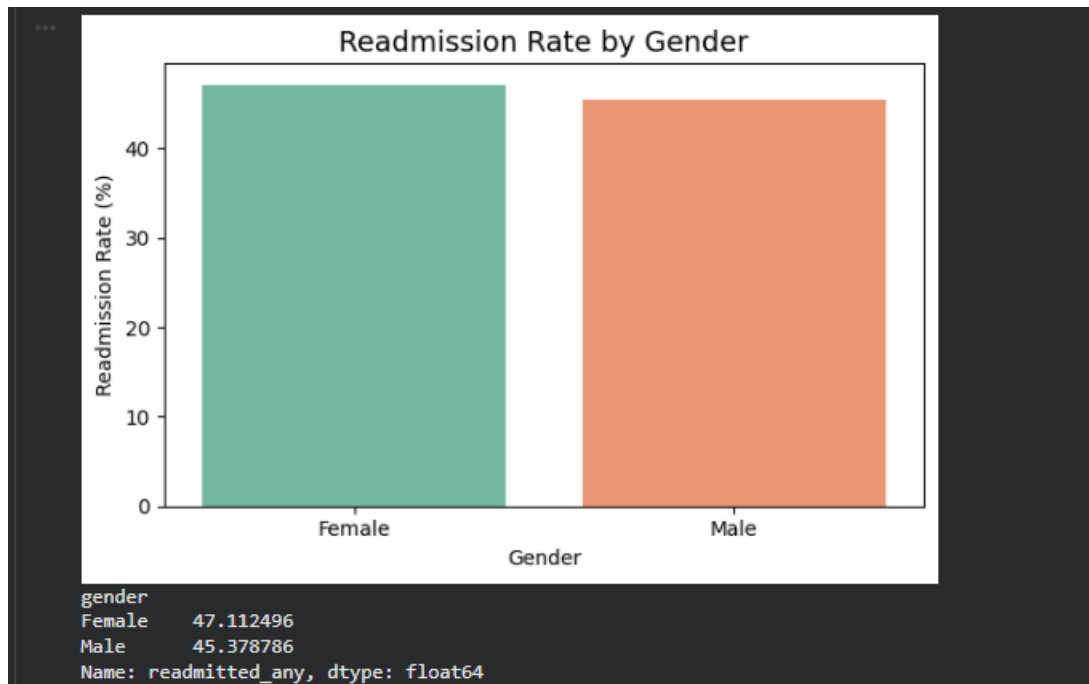
4.2 Age vs Readmission

- Readmission rate increases consistently with age
- 70-80 and 80-90 age groups have the highest rates at 48.2% and 48.3%
- Children aged 0-10 have the lowest readmission rate at 21.5%



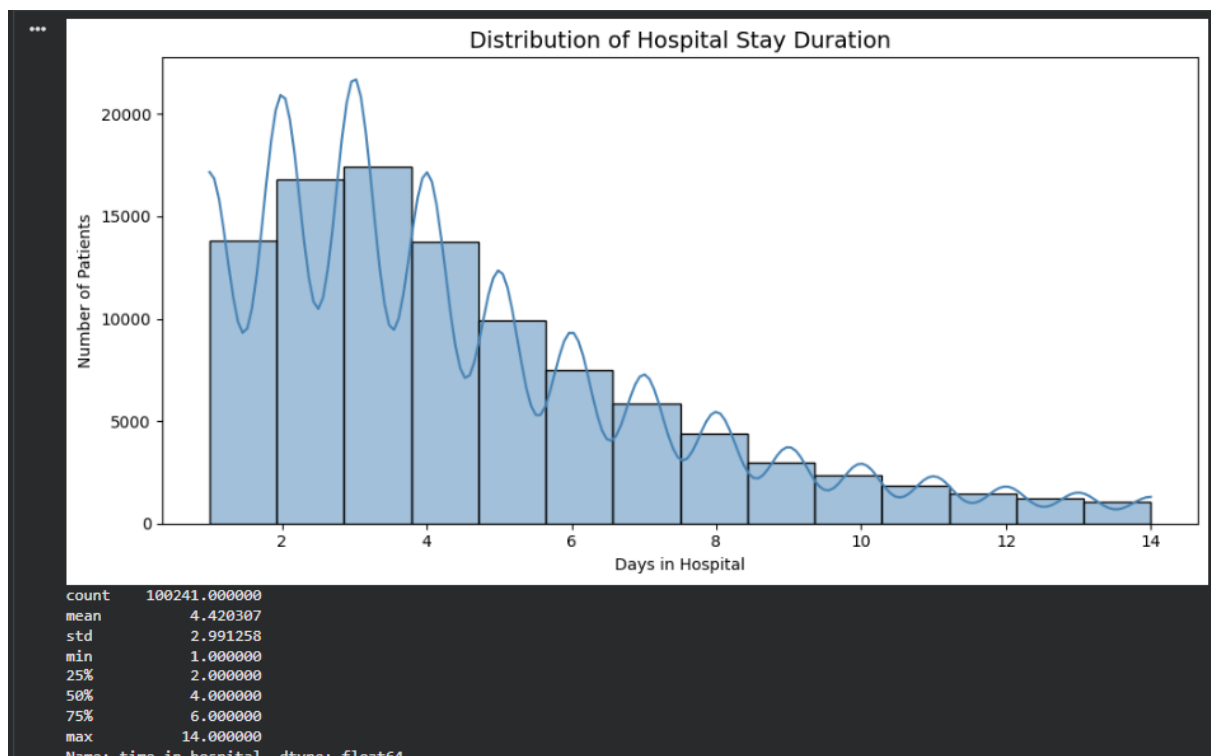
4.3 Gender vs Readmission

- Female patients have slightly higher readmission rate at 47.1% vs Male at 45.4%
- Difference is small gender alone is not a strong predictor



4.4 Hospital Stay Duration

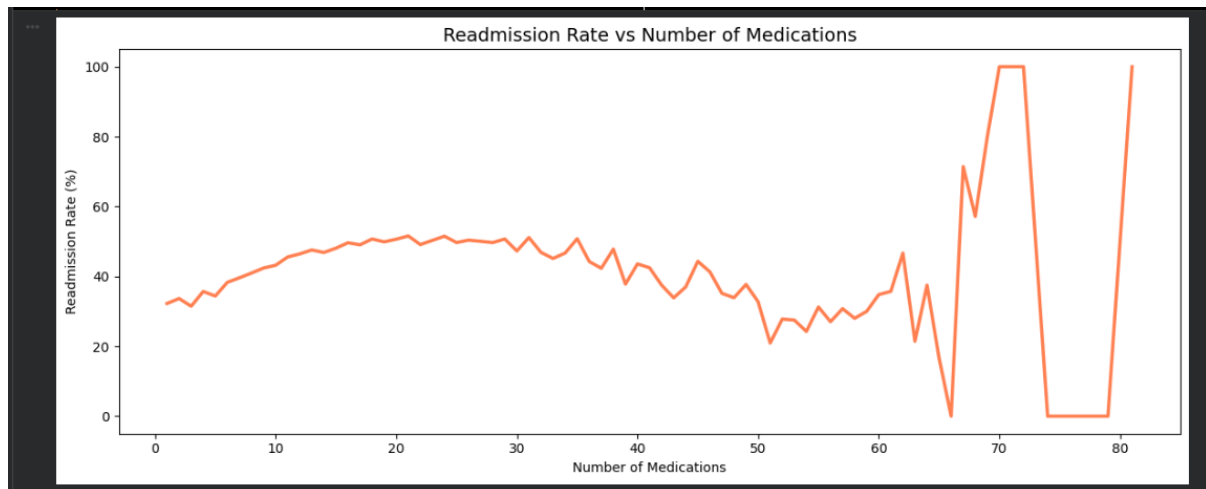
- Average hospital stay is 4.42 days with median of 4 days
- Distribution is right skewed majority of patients stay 1-4 days
- Very few patients stay beyond 10 days



4.5 Medications vs Readmission

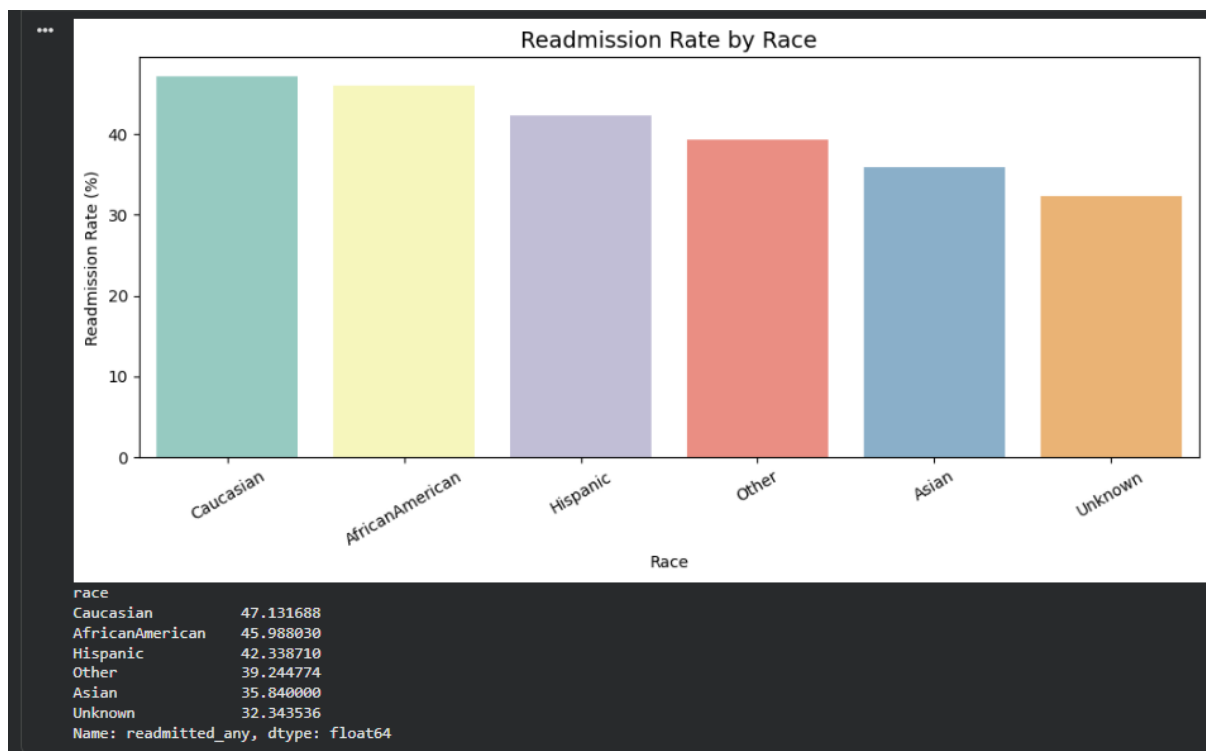
- Clear upward trend — patients on more medications have higher readmission rates

- Patients on 10-30 medications show 45-50% readmission rate
- Spikes beyond 60 medications are unreliable due to very few patients in that range



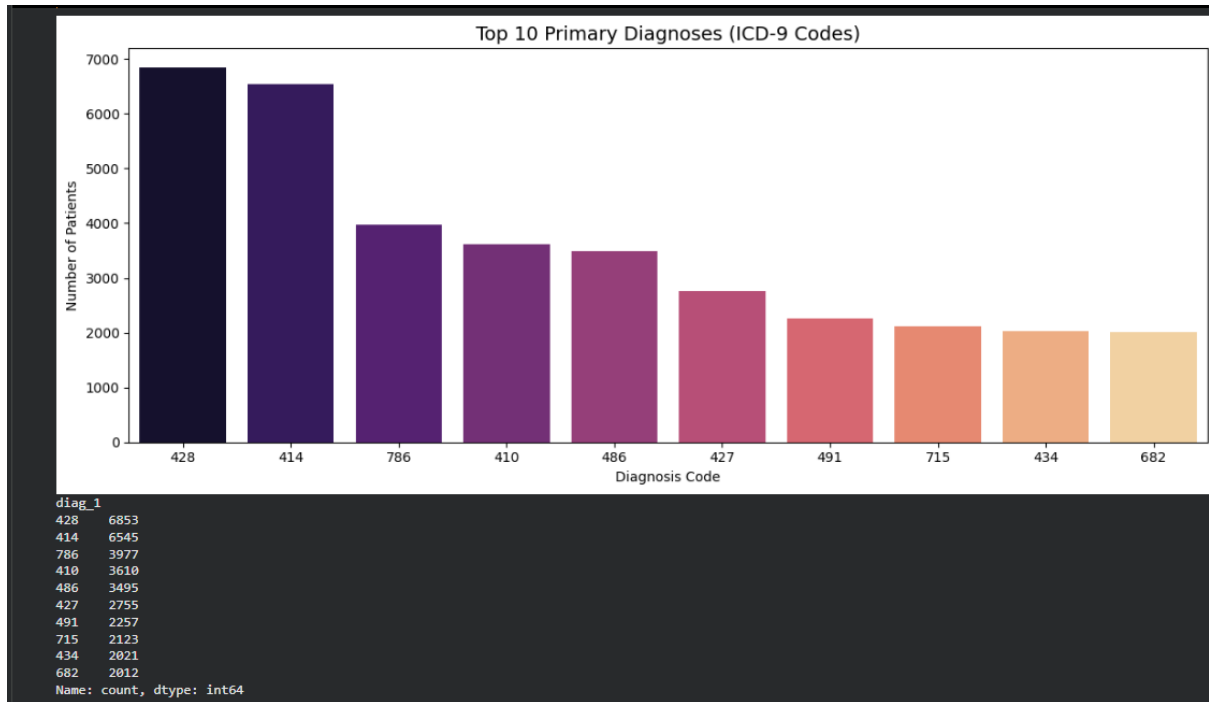
4.6 Race vs Readmission

- Caucasian patients have highest readmission rate at 47.1%
- Asian patients have lowest at 35.8%
- Notable difference across racial groups relevant for equity-focused healthcare planning



4.7 Top Diagnoses

- Heart Failure (ICD-9 code 428) is the most common primary diagnosis with 6,853 patients
- Coronary Artery Disease (ICD-9 code 414) is second with 6,545 patients
- Presence of serious cardiac comorbidities alongside diabetes explains the high overall readmission rates



5. SQL Analysis (Phase 3)

Query 1: Readmission Rate by Age Group

- Grouped all patients by age bracket and calculated readmission percentage
- Result: 80-90 age group has highest readmission rate at 48.26%, children 0-10 have lowest at 21.54%

age	total_patients	total_readmitted	readmission_rate_pct
[90-100)	2785	1114	40
[80-90)	17090	8248	48.26
[70-80)	25896	12484	48.21
[60-70)	22287	10336	46.38
[20-30)	1515	684	45.15
[40-50)	9474	4245	44.81
[50-60)	17044	7538	44.23
[30-40)	3616	1564	43.25
[10-20)	469	196	41.79
[0-10)	65	14	21.54

Query 2: Hospital Stay vs Readmission with Moving Average

- Used window function AVG() OVER() with ROWS BETWEEN sliding frame
- Smoothed out day-to-day variation to reveal the underlying trend
- Result: Readmission rate starts at 40% for 1-day stays and crosses 50% from day 8 onwards

time_in_hospital	total_patients	readmitted	readmission_pct	moving_avg_readmission
1	13791	5519	40.02	43.27
2	16808	7498	44.61	44.55
3	17444	7883	45.19	45.25
4	13744	6647	48.36	47.08
5	9899	4759	48.08	48
6	7490	3683	49.17	49.07
7	5831	2869	49.2	49.47
8	4369	2207	50.51	49.88
9	2989	1506	50.38	49.67
10	2338	1172	50.13	49.66
11	1847	889	48.13	48.98
12	1445	710	49.13	48.76
13	1205	568	47.14	48.42
14	1041	513	49.28	48.52

Query 3 : Top 3 Risk Groups per Race using RANK

- Used RANK() OVER (PARTITION BY race) inside a subquery to rank age groups within each race
- Result: Hispanic patients aged 90-100 have highest readmission rate at 52.63%, Caucasian 70-80 group is the largest high risk group with 20,627 patients.

race	age	total_patients	readmission_pct	ranking_order
AfricanAmerican	[70-80)	3793	47.32	1
AfricanAmerican	[60-70)	4202	47.17	2
AfricanAmerican	[40-50)	2739	46.55	3
Asian	[70-80)	178	40.45	1
Asian	[60-70)	156	37.82	2
Asian	[80-90)	90	37.78	3
Caucasian	[70-80)	20627	48.91	1
Caucasian	[80-90)	14519	48.9	2
Caucasian	[20-30)	887	48.48	3
Hispanic	[90-100)	19	52.63	1
Hispanic	[60-70)	482	47.51	2
Hispanic	[70-80)	384	45.31	3
Other	[60-70)	413	43.34	1
Other	[30-40)	82	42.68	2
Other	[70-80)	323	42.41	3
Unknown	[80-90)	388	38.66	1
Unknown	[70-80)	591	36.72	2
Unknown	[10-20)	3	33.33	3

Query 4 : Patient Risk Segmentation using CTE

- Built a two-stage CTE pipeline using CASE WHEN to label patients as Very High Risk, High Risk, or Normal
- Based on three clinical indicators: number of inpatient visits, number of medications, and hospital stay length
- Result: Very High Risk patients have 68.58% readmission rate vs 39.93% for Normal patients.

risk_category	total_patients	total_readmitted	readmission_pct
Very High Risk	3087	2117	68.58
High Risk	25767	15803	61.33
Normal	71387	28503	39.93

Query 5 : Cumulative Readmission Trend

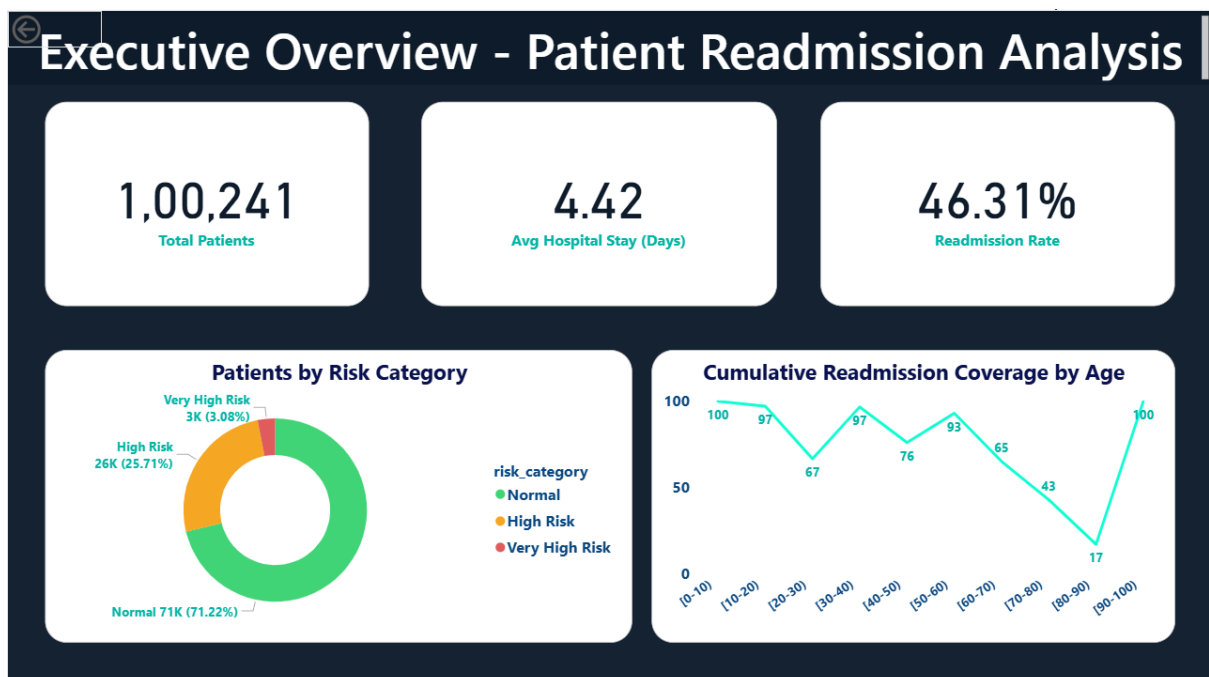
- Used SUM() OVER() running total to calculate cumulative readmission coverage by age group
- Result: Top 3 age groups (60-90) account for 65% of all readmissions confirms Pareto pattern.

age	total_patients	total_readmitted	readmission_pct	cumulative_readmitted	cumulative_patient_pct
[90-100)	2785	1114	40	46409	99.94
[80-90)	17090	8248	48.26	8248	17.05
[70-80)	25896	12484	48.21	20732	42.88
[60-70)	22287	10336	46.38	31068	65.12
[20-30)	1515	684	45.15	31752	66.63
[40-50)	9474	4245	44.81	35997	76.08
[50-60)	17044	7538	44.23	43535	93.08
[30-40)	3616	1564	43.25	45099	96.69
[10-20)	469	196	41.79	45295	97.16
[0-10)	65	14	21.54	46423	100

6. Power BI Dashboard (Phase 4)

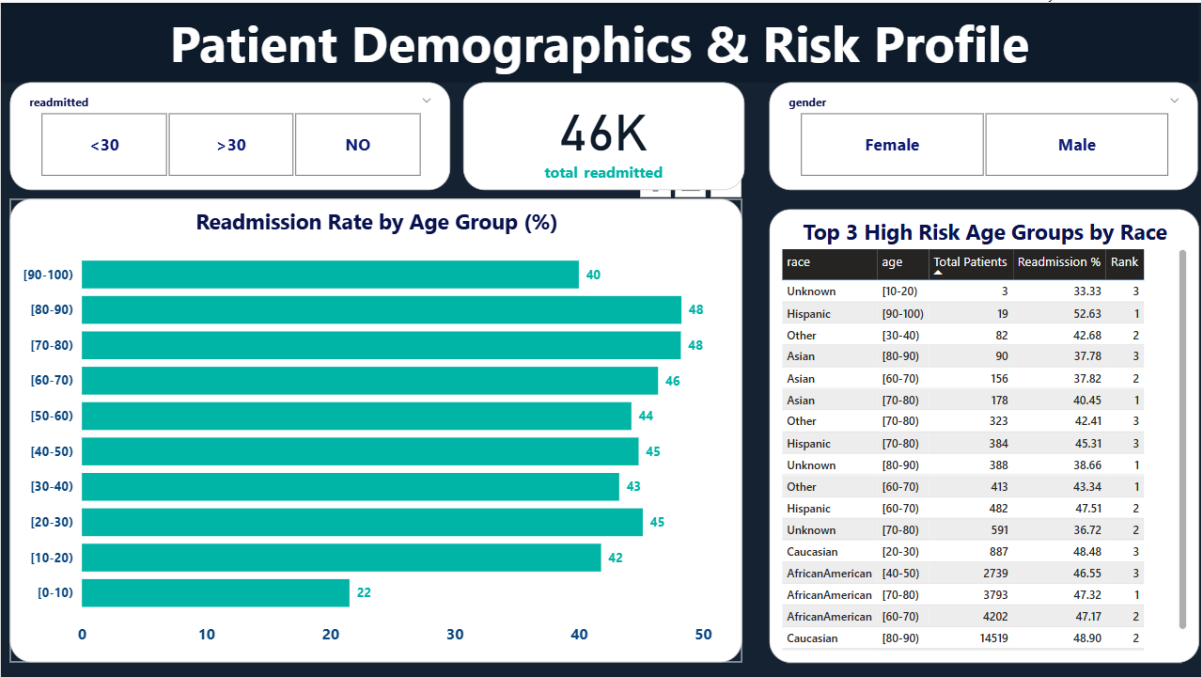
Page 1 : Executive Overview

- Three KPI cards showing Total Patients (100,241), Avg Hospital Stay (4.42 days), Overall Readmission Rate (46.31%)
- Donut chart showing patient distribution by risk category
- Line chart showing cumulative readmission coverage by age group.



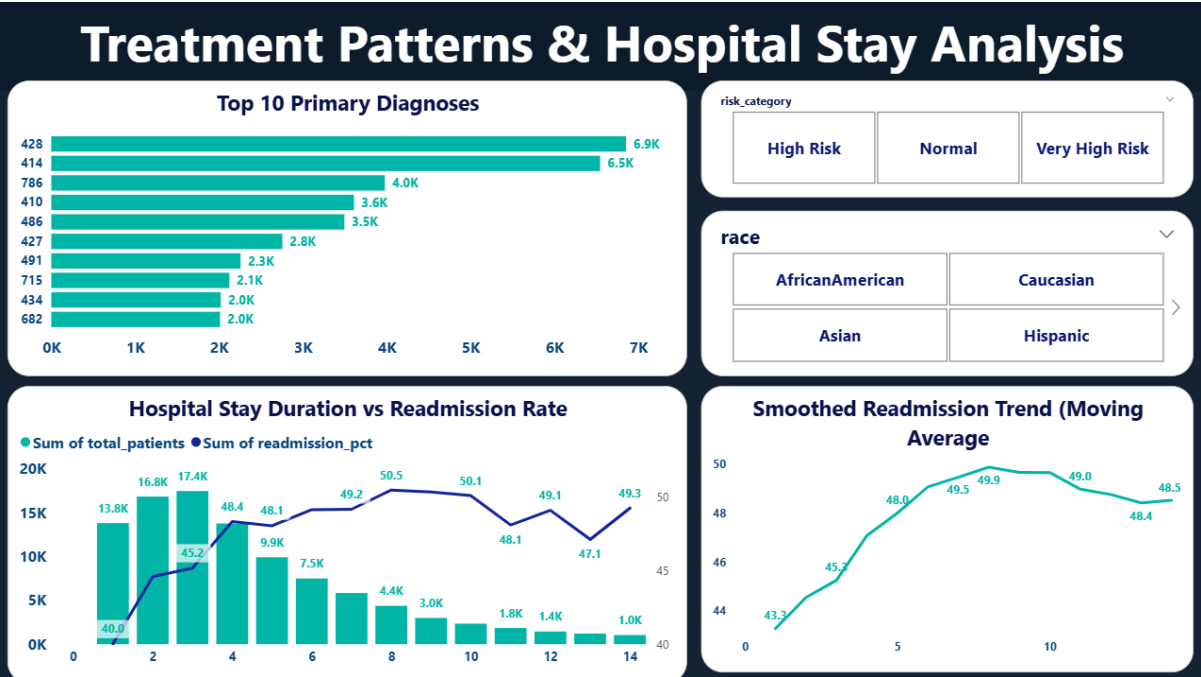
Page 2 : Patient Demographics & Risk Profile

- Horizontal bar chart showing readmission rate by age group
- Matrix table showing top 3 high risk age groups within each racial group
- Interactive slicers for gender and readmission status



Page 3 : Treatment Patterns & Hospital Stay Analysis

- Combo chart showing hospital stay duration against readmission rate and patient volume
- Smoothed moving average trend line for readmission rate
- Top 10 primary diagnoses bar chart
- Interactive slicers for risk category and race



7. Key Findings

- Patients aged 70-90 have nearly double the readmission risk compared to patients under 30
- Very High Risk patients (defined by long stays, many medications, and prior inpatient visits) have a 68.58% readmission rate
- Just the top 3 age groups (60-90) account for 65% of all readmissions a classic Pareto distribution
- Heart Failure and Coronary Artery Disease are the most common comorbidities alongside diabetes
- Readmission rate crosses 50% from day 8 of hospital stay onwards
- Racial disparities exist Caucasian and African American patients have notably higher readmission rates than Asian patients.

8. Business Recommendations

- Target elderly patients aged 70-90 with dedicated post-discharge follow-up and monitoring programs
- Implement early flagging system for Very High Risk patients using the three-factor risk model
- Allocate additional cardiac care resources for Heart Failure and CAD patients as they represent the highest volume diagnoses
- Intervene before day 8 of hospital stay to reduce the risk of long-stay readmissions
- Design race-aware care programs to address the readmission gap between demographic groups

9. Limitations

- Dataset is from 1999-2008 and may not reflect current medical practices, medications, or treatment protocols
- Weight column had 96.8% missing values and had to be dropped weight is a clinically significant factor for diabetes
- A1C result and glucose serum levels were also mostly missing these are key diabetic indicators that could have strengthened the analysis
- Medical specialty column was 49% missing specialty-wise analysis was not possible
- No temporal analysis was done patient readmission patterns over time were not explored.

10. Conclusion

- This project successfully demonstrates an end-to-end healthcare data analytics workflow
- Starting from raw messy data with placeholder missing values, the dataset was cleaned, explored, analyzed with SQL, and visualized in a professional dashboard
- The analysis confirms that age, hospital stay duration, number of medications, and prior inpatient visits are strong indicators of readmission risk
- The Power BI dashboard provides an interactive tool that hospital administrators could use for resource planning and patient prioritization
- Advanced SQL concepts including window functions, CTEs, and subqueries were used to answer real business questions beyond basic aggregations
- This analysis lays the foundation for a predictive machine learning model that could automate readmission risk scoring in a clinical setting
- The risk segmentation model (Very High / High / Normal) uses fixed thresholds defined manually a machine learning model could produce more accurate segmentation.